

Essentials of Data Science
Laboratory - 2304102L - 2026

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4.2.8. Titanic Dataset Analysis and Data Cleaning - 4 00:36

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked_S).
4. Get the number of non-survivors by embarkation location (Embarked_S).
5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below,

P	S	P	N	S	A	S	P	T	F	C	E
a	u	l	a	e	g	i	a	i	a	a	m
s	r	a	m	x	e	b	r	c	r	b	b
s	v	s	e			S	h	k	e	i	
e	i						e		n		
n	v								k		
g	e								e		
i	d								d		

titanicDat... Submit

```

1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data = pd.get_dummies(data, columns=['Embarked'],
7 drop_first=True)
8
9 # 1. Get the number of survivors by gender
10
    
```

Average time: 0.082 s (82.00 ms) Maximum time: 0.082 s (82.00 ms) 1 out of 1 shown test case(s) passed

Test case 1 (82 ms) Debug

Expected output	Actual output
female...233	female...233
male...109	male...109
Name: Sex, dtype: int64	Name: Sex, dtype: int64
male...468	male...468
female...81	female...81
Name: Sex, dtype: int64	Name: Sex, dtype: int64

Terminal Test cases

4.1. Practice Lab Assignment

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5. Practical 5

4.2.7. Titanic Dataset Analysis and Data Cleaning - 3

00:48

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked_S).
3. Calculate the survival rate by family size (FamilySize).
4. Calculate the survival rate by being alone (IsAlone).
5. Get the average fare by passenger class (Pclass).
6. Get the average age by passenger class (Pclass).
7. Get the average age by survival status (Survived).
8. Get the average fare by survival status (Survived).
9. Get the number of survivors by class (Pclass).
10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

P a s s e n g e r I d	S u r v i v e d	P l a s s	N a m e	S e x	A g e	S i b S p	P a r c h	T i c k e t	F a r e	C a b i n	E m b a r k e d
---	--------------------------------------	-----------------------	------------------	-------------	-------------	-----------------------	-----------------------	----------------------------	------------------	-----------------------	--------------------------------------

titanicDat...

Submit

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data['FamilySize'] = data['SibSp'] + data['Parch']
7 data['IsAlone'] = np.where(data['FamilySize'] > 0, 0,
8                             1)
9 data = pd.get_dummies(data, columns=['Embarked'],
                             drop_first=True)
```

Average time

0.090 s
90.00 ms

Maximum time

0.090 s
90.00 ms

1 out of 1 shown test case(s) passed

Test case 1

90 ms

Debug

Expected output

Actual output

Pclass

Pclass

1...0.629630

1...0.629630

2...0.472826

2...0.472826

3...0.242363

3...0.242363

Name: Survived, dtype: float64

Name: Survived, dtype: float64

Embarked_S

Embarked_S

Terminal

Test cases

4.1. Practice Lab Assignment

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5. Practical 5

4.2.6. Titanic Dataset Analysis and Data Cleaning - 2

03:01

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
2. Convert the 'Sex' column to numeric values (male: 0, female: 1).
3. One-hot encode the 'Embarked' column, dropping the first category.
4. Get the mean age of passengers.
5. Get the median fare of passengers.
6. Get the number of passengers by class.
7. Get the number of passengers by gender.
8. Get the number of passengers by survival status.
9. Calculate the survival rate of passengers.
10. Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

P a s s e n g e r I d	S u r v i v e d	P l a s s	N a m e	S e x	A g e	S i b S p	P a r c h	T i c k e t	F a r e	C a b i n	E m b a r k e d
---	--------------------------------------	-----------------------	------------------	-------------	-------------	-----------------------	-----------------------	----------------------------	------------------	-----------------------	--------------------------------------

titanicDat...

Submit

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data['FamilySize'] = data['SibSp'] + data['Parch']
7
8 '''# 1. Create a new column 'IsAlone' (1 if alone, 0
9 otherwise)
10
```

Average time

0.075 s
75.00 ms

Maximum time

0.075 s
75.00 ms

1 out of 1 shown test case(s) passed

Test case 1 75 ms

Debug

Expected output

Actual output

29.69911764705882

29.69911764705882

14.4542

14.4542

3...491

3...491

1...216

1...216

2...184

2...184

Name: Pclass, dtype: int64

Name: Pclass, dtype: int64

Terminal

Test cases

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4.1. Practice Lab Assignment

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5. Practical 5

4.2.5. Titanic Dataset Analysis and Data Cleaning

01:31

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.
2. Display the last 5 rows of the dataset.
3. Get the shape of the dataset (number of rows and columns).
4. Get a summary of the dataset (using `.info()`).
5. Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.
6. Check for missing values and display the count of missing values for each column.
7. Fill missing values in the 'Age' column with the median age.
8. Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).
9. Drop the 'Cabin' column due to many missing values.
10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

P	S	P	N	S	A	S	P	T	F	C	E
a	s	l									m
s	s										b

Explorer titanicDat...

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 '''# 1. Display the first 5 rows of the dataset
8
9
10 # 2. Display the last 5 rows of the dataset
11
```

Average time 0.255 s 255.00 ms Maximum time 0.255 s 255.00 ms 1 out of 1 shown test case(s) passed

Test case 1 255 ms Debug

Expected output	Actual output
PassengerId Survived Pclass Fare Cabin Embarked	PassengerId Survived Pclass Fare Cabin Embarked
0 1 0 3 7.2500 NaN S	0 1 0 3 7.2500 NaN S
1 2 1 1 71.2833 C85 C	1 2 1 1 71.2833 C85 C
2 3 1 3 7.9250 NaN S	2 3 1 3 7.9250 NaN S

Terminal Test cases

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4.2.4. Most Frequently Sold Product Pairs

00:58

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pair/s that was sold most frequently.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Explanation:

Transactions:

- 2025-01-01: Product A, Product B
- 2025-01-02: Product A, Product C
- 2025-01-03: Product B, Product A
- 2025-01-04: Product C, Product B

Explorer

frequentl...

```
1 import pandas as pd
2 from itertools import combinations
3 from collections import Counter
4
5 # Prompt user to input the file name
6 file_name = input()
7
8 # Read data from the specified CSV file
9 df = pd.read_csv(file_name)
10
11 # write the code
```

Average time

0.019 s

19.33 ms

Maximum time

0.036 s

36.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 36 ms

Debug

Expected output

sales_data.csv

Product A and Product B: 2 times

Product A and Product C: 2 times

Actual output

sales_data.csv

Product A and Product B: 2 times

Product A and Product C: 2 times

Terminal

Test cases

Course

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5. Practical 5

4.2.3. City that Sold the Most Products

00:38

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the columns: Date, Product, Quantity, Price, and City.

Group the data by city and calculate the total quantity of products sold for each city.

Find the city that sold the most products (based on the total quantity sold).

Sample Data:

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New York

2025-01-02,Product C,4,30,Chicago

2025-01-03,Product B,2,15,Chicago

2025-01-03,Product A,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product B,5,15,Los Angeles

2025-01-05,Product A,3,20,Chicago

2025-01-05,Product C,10,30,Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

monthFor...

Submit

1import pandas as pd

2

3# Prompt the user for the file name

4file_name = input()

5

6# Load the data

7df = pd.read_csv(file_name)

8

9# write the code..

10city_sales=df.groupby('City')['Quantity'].sum()

11best_city=city_sales.idxmax()

Average time

0.019 s

19.33 ms

Maximum time

0.038 s

38.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 138 msDebug

Expected output

sales_data.csv

City sold the most products: Los Angeles

Actual output

sales_data.csv

City sold the most products: Los Angeles

TerminalTest cases

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4.2.2. Best Selling Product

15.40

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

monthFor...

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8
9
10 # Find the product with the highest total quantity sold
```

Average time 0.028 s 27.67 ms Maximum time 0.059 s 59.00 ms

1 out of 1 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 59 ms Debug

Expected output	Actual output
sales_data.csv	sales_data.csv
Best-selling-product: Product-A	Best-selling-product: Product-A
Total-quantity-sold: 23	Total-quantity-sold: 23

Terminal Test cases

Course

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4. Practical 4

4.1. Practice Lab Assignment

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4.2.1. Month with the Highest Total Sales

01:04

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the columns: Date, Product, Quantity, Price, and City.

Group the data by Month and calculate the total sales for each month.

Find the month with the highest total sales and display it.

Also, display the total sales for the best month.

Sample Data:

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New York

2025-01-02,Product C,4,30,Chicago

2025-01-03,Product B,2,15,Chicago

2025-01-03,Product A,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product B,5,15,Los Angeles

2025-01-05,Product A,3,20,Chicago

2025-01-05,Product C,10,30,Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

monthFor...

Submit

1 import pandas as pd

2

3 # Prompt the user for the file name

4 file_name = input()

5

6 # Load the data

7 df = pd.read_csv(file_name)

8

9

10 # Find the month with the highest total sales

11 df = pd.read_csv(file_name)

Average time 0.029 s 29.33 ms

Maximum time 0.058 s 58.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 58 ms Debug

Expected output sales_data.csv

Actual output sales_data.csv

Best-month: 2025-01

Total-sales: \$1210.00

Best-month: 2025-01

Total-sales: \$1210.00

Terminal Test cases

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- 3.2.1. Numpy: Matrix Operations
- 3.2.2. Numpy: Horizontal and Vertical Stac...
- 3.2.3. Numpy: Custom Sequence Generation
- 3.2.4. Numpy: Arithmetic and Statistical Op...
- 3.2.5. Numpy: Copying and Viewing Arrays
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4. Practical 4

4.1. Practice Lab Assignment

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4.1.3. Student Information

31:50

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:

Refer to the displayed test cases for better understanding.

Sample Test Cases

Explorer

studentin...

Submit

```
1 import pandas as pd
2
3 # Read the text file into a DataFrame
4 file = input()
5 data = pd.read_csv(file, sep="\s+", header=None,
6 names=["Name", "Age", "Grade"])
7
8 # write your code here..
9
10 print("First five rows:")
```

Average time

0.041 s
41.00 ms

Maximum time

0.060 s
60.00 ms

1 out of 1 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Test case 1 60 ms

Debug

Test cases

Expected output

studentdata.txt

First five rows:

Name Age Grade

0 John 25 A

1 Alin 22 B

2 Emma 24 A

Actual output

studentdata.txt

First five rows:

Name Age Grade

0 John 25 A

1 Alin 22 B

2 Emma 24 A

Terminal

Test cases

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- 3.2.1. Numpy: Matrix Operations
- 3.2.2. Numpy: Horizontal and Vertical Stac...
- 3.2.3. Numpy: Custom Sequence Generation
- 3.2.4. Numpy: Arithmetic and Statistical Op...
- 3.2.5. Numpy: Copying and Viewing Arrays
- 3.2.6. Numpy: Searching, Sorting, Countin...
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4. Practical 4

4.1. Practice Lab Assignment

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5. Practical 5

4.1.2. Dictionary to dataframe

01:31

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column **Gender** with values taken from the user.
- Display the DataFrame after adding the new column.

datafram... Submit

```

1  '''import pandas as pd
2
3  # Provided dictionary of lists
4  data = {
5      'Name': ['Alice', 'Bob', 'Charlie'],
6      'Age': [25, 30, 35],
7  }
8
9  # Convert the dictionary to a DataFrame
10 df = pd.DataFrame(data)
11

```

Average time
0.085 s
84.50 ms

Maximum time
0.090 s
90.00 ms

1 out of 1 shown test case(s) passe
1 out of 1 hidden test case(s) passe

Test case 1 90 ms Debug

Expected output

Original DataFrame:

.....Name Age

0.....Alice.....25

1.....Bob.....30

2.....Charlie.....35

New name: Susan

Actual output

Original DataFrame:

.....Name Age

0.....Alice.....25

1.....Bob.....30

2.....Charlie.....35

New name: Susan

Terminal Test cases

Course

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3.2.1. Numpy: Matrix Operations

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4.1.1. Pandas - series creation and manipulation

07:31

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the `groupby` and `mean()` operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

The user should enter a list of numbers separated by space when prompted.

Output Format:

The program should display the mean of even and odd numbers separately.

Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

Sample Test Cases

seriesMa...

Submit

1 import pandas as pd

2

3 # Take inputs from the user to create a list of numbers

4 numbers = list(map(int, input().split()))

5

6 # Create a Pandas series from the list of numbers

7 series=pd.Series(numbers)

8 # Grouping by even and odd numbers and calculating the mean

9 grouped = series.groupby(series%2==0).mean()

Average time

0.014 s

14.17 ms

Maximum time

0.049 s

49.00 ms

3 out of 3 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 12 ms

Debug

Expected output

1 2 3 4 5 6 7 8 9 10

Mean of even and odd numbers:

Odd: 5.0

Even: 6.0

dtype: float64

Actual output

1 2 3 4 5 6 7 8 9 10

Mean of even and odd numbers:

Odd: 5.0

Even: 6.0

dtype: float64

Terminal

Test cases